

### 5.1.2 Solid, liquid and gas



| Learning outcomes                                                                                                                            | Additional guidance |
|----------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>                                                |                     |
| <b>(a)</b> solids, liquids and gases in terms of the spacing, ordering and motion of atoms or molecules                                      |                     |
| <b>(b)</b> simple kinetic model for solids, liquids and gases                                                                                |                     |
| <b>(c)</b> Brownian motion in terms of the kinetic model of matter and a simple demonstration using smoke particles suspended in air         |                     |
| <b>(d)</b> internal energy as the sum of the random distribution of kinetic and potential energies associated with the molecules of a system |                     |

- (e)** absolute zero (0 K) as the lowest limit for temperature; the temperature at which a substance has minimum internal energy
- (f)** increase in the internal energy of a body as its temperature rises
- (g)** changes in the internal energy of a substance during change of phase; constant temperature during change of phase.